Getting started with spreadsheets.

1. Basic Computation

   (a) To add 3 + 4, enter `=3+4`

   (b) To subtract 100-76, enter `=100-76`

   (c) To multiply 4 times 18, enter `=4*18`

   (d) To divide 0.05 by 12, enter `=0.05/12`

   (e) To calculate $5^{25}$, enter `=5^25`

| | A | B |
|---|---|---|
| 1 | enter into a column with a leading = | answer |
| 2 | 3+4 | 7 |
| 3 | 100-76 | 24 |
| 4 | 4*18 | 72 |
| 5 | 0.05/12 | 0.004166666667 |
| 6 | 5^25 | 2.98023E+17 |
| 7 | | |

2. Compute an 18% tip on a $35.75 bill.

   (a) What is 18% as a decimal? 18/100 = 0.18

   (b) What should we enter into the spreadsheet? 35.75*0.18 ≈ $6.44

3. Make a tip calculator using **cell references**.

   (a) Open a new spreadsheet and label it `TIP`.

   (b) In cells A1, B1, C1, D1, type column headers:

   | `Bill Amount` | `Tip Percent` | `Tip Paid` | `Total` |
   |---|---|---|---|

   (c) In cells A2, B2, C2, D2, type:

   | $= 35.75$ | $= 0.18$ | $= A2 * B2$ | $= A2 + C2$ |
   |---|---|---|---|

   (d) What calculation did we do?

   C2 calculates the tip

   D2 calculates the total

   (e) Use the calculator to determine the Total if the Bill Amount is $76.80 and the Tip Percent is 20%.

   Change A2 to 76.80 & B2 to 0.2

   total = $92.16

4. Keeping the Tip Percent at 20%, we now demonstrate the **Fill Down** feature.

   (a) Enter 10 into cell `A2` and enter `=A2+10` into cell `A3`

   (b) Select `A3` and drag down until you get to 110.

   (c) Drag down cells `B2` through `D2`.

   (d) How much is the tip on a $110 meal? $22

   How much is the final bill? $132

   How large does the Bill Amount need to be for the tip to be more than $15? $80

   (the tip is $16)

| | A | B | C | D |
|---|---|---|---|---|
| 1 | Bill Amount | Tip Percent | Tip Paid | Total |
| 2 | $10.00 | 0.2 | $2.00 | $12.00 |
| 3 | $20.00 | 0.2 | $4.00 | $24.00 |
| 4 | $30.00 | 0.2 | $6.00 | $36.00 |
| 5 | $40.00 | 0.2 | $8.00 | $48.00 |
| 6 | $50.00 | 0.2 | $10.00 | $60.00 |
| 7 | $60.00 | 0.2 | $12.00 | $72.00 |
| 8 | $70.00 | 0.2 | $14.00 | $84.00 |
| 9 | $80.00 | 0.2 | $16.00 | $96.00 |
| 10 | $90.00 | 0.2 | $18.00 | $108.00 |
| 11 | $100.00 | 0.2 | $20.00 | $120.00 |
| 12 | $110.00 | 0.2 | $22.00 | $132.00 |

5. Quick Summary of Finance Terms

(a) **principal**

Starting amount of money

also called present value

(b) **interest rate**

r = a percent that determines what percent of the principal should be added as interest

(c) **interest**

an amount of money added in each time period

(d) **accumulated amount** or **end amount** or **principal plus interest** or **future value**

$A = P + I$ = starting amount plus interest added

(e) **Formula for Simple Interest over Time**

$A = P + I = P + Prt = P(1+rt)$

P: principal; I: interest; Prt: how to calculate interest; P(1+rt): same, by algebra!

(f) Comment on time $t$.

interest can be added in using different time periods: annually, quarterly, monthly, etc.

6. Make a Simple Interest Calculator:

Suppose $1000 is invested in a savings account that earns 3.2% interest on the $1000 each year.

↑ principal ↑ interest rate

(a) Open a new spreadsheet/tab called `Interest`. Fill in the spreadsheet cells:

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | Interest Rate | Principal | Year | Interest | Total |
| 2 | 0.032 | 1000 | 0 | | =B2+D2 |
| 3 | 0.032 | 1000 | =C2+1 | =A3*B3 | =D3+E2 |

(b) Use the pull down feature to determine the Total in the savings account in 10 years. Write down your steps.

Select A3 thru E3 & pull down until year = 10

(c) How long will it take for the savings account to double in value? Write down your steps.

We need total > 2000

Keep dragging! Eventually we see total = 2024 in year 32

(d) How can you calculate the Total in the savings account in 50 years **without** using the pull down feature?

If $P$ is the principal, $r$ is the interest rate, $t$ is the number of years, and $A$ is the accumulated amount of money in the savings account, write a formula for $A$. Then check your calculation using the pull down feature.

(e) A better simple interest calculator, using fixed cell references (the $ fixes the row/column)

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | Interest Rate | Principal | Year | Interest | Total |
| 2 | 0.032 | 1000 | 0 | | |
| 3 | | | =C2+1 | = $A$2*$B$2 | =D3+E2 |

Now select and pull down `C3, D3, E3`. What happens if you change the interest rate to 4%? What if you change the principal to $3000?

Everything updates w/o re-pulling!

7. How To Export the **entire workbook** as a **PDF**.

(a) Save your work!

(b) In upper left menu, click `File` to produce a dropdown menu.

(c) In dropdown menu, select `Download`

(d) In right-side menu, select `PDF`